Newton, quantum mechanics and Dimock–Balaban: a guide through one time dimension

Scope and result, 2026-10-27. A point particle already displays several different questions hidden in the phrase "existence of motion": does a trajectory exist, is it unique, does it continue for all time, and what rule applies at a wall or singularity? This guide connects those questions to a constructive $0+1$ quantum theory and to the multiscale organization of Dimock's exposition of Bałaban's renormalization group. The simple model is not a solution of the four-dimensional Yang–Mills problem.

The opening synthesis collects the recent results and their proof links. Sections 1–8 explain the constructions, their sources and the assumptions needed for classical examples. Source-reading labels distinguish cited statements from analogies or calculations made here; no priority claim is made for standard ODE, Gaussian, semigroup or semiclassical methods.

Recent Newton synthesis

This section consolidates the work of **27–28 October 2026**. The individual notes remain the proof records: their hypotheses, source reading labels, corrections and checks govern the statements below. The newer calculations are written derivations, some with targeted internal model checks; they are not independent peer review. Neither the existence of a quantum construction nor any conditional floor here establishes that a nonzero physical $\hbar$ is necessary for every consistent completion of Newtonian mechanics.

We keep physical $\hbar$ explicit. A formal classical limit may be written $\epsilon = \hbar/S_0 \to 0$ for a specified characteristic action S_0; the deformation parameter ϵ, the time mesh and the resolution of a record can be related by a specified control map. If physical $\hbar$ is held fixed, the first limit requires a family with increasing characteristic action, not merely a change of units.

0. Groupoid scale and renormalization

The connection is already part of Rivero's work. The appendix to his 1997 tangent-groupoid note (pp. 7–8, passages read) constructs a **toy RG triangle**: a dilation rescales a bare sequence approaching the tangent boundary to a chosen nonzero reference slice, with limits at different slices related by rescalings. His 1998 Feynman-formula proposal (pp. 2–3, equations (5)–(10), passages read) similarly distinguishes two regulators and calls its controlled objects bare, renormalized and dressed series. Its control transformation is left unspecified; its control constant is arbitrary. The 1999 strict-quantization paper with Rivero and coauthors supplies the groupoid quantization framework. These are the provenance of the present question, not a claim that the recent calculation invented the groupoid/RG connection.

The different appearances of epsilon need an explicit dictionary. Let ϵ_G be a dimensionless groupoid deformation parameter; in the specified state formulation, choosing $\epsilon_G = \hbar/S_0$ assigns the dimensionful phase constant to that parameter. Let $\eta = |\pi|/T_0$ be a dimensionless mesh scale, and let α_π be an action-dimensional bare phase regulator. A possible control map, already recorded in the 1998 test, is

$$S_\pi^{\mathrm{bare}} = Z_S(\eta) S_\pi, \qquad \hbar_{\mathrm{eff}} = \frac{\alpha_\pi}{Z_S(\eta)}.$$

For example, the *specified* choices $Z_S(\eta) = \eta$ and $\alpha_\pi = \eta\hbar$ give $e^{iS_\pi^{\mathrm{bare}}/\alpha_\pi} = e^{iS_\pi/\hbar}$ at every mesh. This algebra illustrates how a vanishing bare regulator can coexist with a fixed positive effective action; substituting the same regulator into an unscaled action would give a different phase.

An exact Gaussian blocking test. Let $m > 0$, $u \in \mathbb{R}$ and let $a \geq 0$ be an action-dimensional parameter. Define a normalized transition kernel by

$$K_{a,t}^u(x, dy) = \mathcal{N}(x + ut, at/m)(dy), \qquad t > 0,$$

with zero variance interpreted as a Dirac measure. For $a = \hbar > 0$ and $u = 0$ this is the free Euclidean quantum kernel; arbitrary u gives the drifted Gaussian reference, not a real-time path measure. Integrating out an intermediate position preserves a exactly:

$$K^u_{a,t} K^u_{a,v} = K^u_{a,t+v}.$$

If time and position are then rescaled by $t_{\text{old}} = r t_{\text{new}}$, $x_{\text{old}} = s x_{\text{new}}$ with $r, s > 0$, the normalized kernel obeys

$$K^u_{a,rt}(sx, sB) = K^{ru/s}_{ra/s^2,t}(x, B) \quad \text{for every Borel set } B.$$

These identities follow from adding independent Gaussian increments and rescaling their means and variances; they include $a = 0$. Under the diffusive choice $r = b$, $s = \sqrt{b}$, the action coefficient stays fixed. When $u = 0$, every $a \geq 0$ is a fixed point of this particular normalized block-and-rescale map. At $a = 0$, retaining u gives the deterministic free Newton motion $x(t) = x(0) + ut$. Thus this exact free RG control neither selects a positive action nor excludes the classical branch. It is an illustrative Gaussian comparison, not a result about all interacting blocking maps or complete record instruments; compare the free increment and the action-normalization identity.

The open RG problem is to construct a compatible control or blocking map for the actual normalized kernels, states and generated interactions, including the increments and all required limits, and then determine whether its effective action has a universal physical identification. Geometric zoom, coarse-graining and running parameters can participate in one construction, but a geometric dilation alone does not establish that construction or force $\hbar$ to run. The corrected groupoid comparison and scale correspondence keep these claims and their corrections distinct.

1. Refinement builds motion; it does not select an action unit

For a globally Lipschitz acceleration with constant Λ, the causal kick–drift polygons on any positive partitions, even with the order chosen independently in each cell, converge as the maximum cell length tends to zero to the unique initial-value solution. Their stability constants do not depend on the number of cells (ODE lesson, §§1–3). The separate two-endpoint problem has a unique polygon and a unique continuous solution under the sufficient condition $\Lambda T^2/8 < 1$; this is a short-time boundary-value condition, not a restriction on causal continuation (time blocking, §2).

Lipschitz regularity is sufficient, not necessary. In the specified repelling logarithmic family, forces behaving near zero like $q[\log(\ell/|q|)]^\beta$ have a unique resting solution for $0 < \beta \leq 2$, but permit waiting and departures for $\beta > 2$. The zero-energy arrival-time integral distinguishes the cases; action on shrinking annuli tends to zero on both sides (time scales, §§1–4).

One normalized Euclidean time cell with a convex potential gives explicit fluctuation and effective-curvature bounds. At a quartic flat centre its positive-action integral generates curvature while classical minimization has zero quadratic curvature. This is a consequence of a supplied positive weight, not a selection of its action multiplier (nonlinear cell, §§2–5). The Alpoge–Sussman comparison records a fourth-response and reflection-positivity check in $0 + 1$; it imports no four-dimensional Yang–Mills proof.

2. Conditional routes to a floor

Response ceiling. In the specified double-well Euclidean cell, write $a = m(u + v)/(uv)$ and let $r > 0$ be the location of the two symmetric minima. If the physical normalized response has exactly the stated positive-weight form with action multiplier $\eta > 0$, and an independently fixed finite ceiling $M'_\eta(0) \leq R_*$ holds, then

$$\eta > \frac{ar^2}{3R_*} > 0.$$

The particular zero-action minimization rule has a branch jump, but a classical retained branch label avoids that jump and a classical statistical weight can reproduce the response. The response law, ceiling and identification of η with physical $\hbar$ are additional premises (branch-response theorem, §§2–5).

Cusp and reversible phase states. A bounded continuous force has two distinct point trajectories that merge in finite time. Let a nonempty preparation class have a finite-covariance member, be invariant under affine canonical transformations and weakly closed, and admit reversible evolution with weakly continuous state curves, compatible with the incoming point trajectories when those points are admitted. Then the infimum of

$$\kappa(\mu) = \sqrt{\det \mathrm{Cov}_\mu(q,p)}$$

over its finite-covariance preparations is strictly positive (cusp floor, §2). Compatibility can be vacuous when no point state is admitted. A classical density-ceiling model with actual almost-everywhere cusp flow realizes the premises and has a supplied scale; it fails closure under a two-copy canonical mixer.

Continuity instead of reversibility. A later theorem assumes a W_2-closed affine-canonical class with a positive-definite covariance member, independent-copy orthogonal canonical mixers and retained marginals. At one specified sufficiently short positive time, require a map continuous from W_2 inputs to weak outputs that agrees with the actual cusp flow on every admitted nondegenerate Gaussian. Its covariance-action infimum is positive even without injectivity, a semigroup law or state-space invariance of the output (history and continuity, Theorem 3). An entropy-floor classical class realizes these assumptions with its scale imposed. Neither floor is identified with $\hbar/2$.

With affine-canonical invariance, independent copies, canonical mixing, marginalization and either weak or W_2 closure also admit the Gaussian with each admissible mean and covariance. If a positive covariance infimum is supplied or separately proved, the entire Gaussian sector above that floor follows; this does not determine its non-Gaussian sector or make it quantum (canonical central limit, §§2–4).

3. Classical histories and complete records change the question

The cusp admits a reversible, time-reversal-symmetric classical group on the space of **complete trajectories**, with Dirac states and classical conditioning. Its instantaneous phase point does not determine the shifted trajectory: the enlarged state retains future-selection information. No finite-dimensional local symplectic apparatus realizing that access is constructed (history model, Theorem 1).

An earlier three-body spring calculation gives a distinct, finite-dimensional Hamiltonian example: eliminating the receiver leaves reversible memory, and two nearby exact tagged-phase observations recover its hidden state. This makes state loss at a cut concrete; it does not realize the cusp force or select cusp histories.

In calibrated variables, take independent standard Gaussians X, Y and initial data $Q_0 = \delta(u + X)$, $P_0 = \delta Y$. As $\delta \to 0$, all initial laws tend to the same point, but their laws on a fixed short positive-time interval tend to a mixture of the two departing cusp paths with weights $\Phi(u)$ and $\Phi(-u)$. Thus actual classical phase-law transport cannot be weakly continuous at that point for all these preparations; these are classical branch probabilities, not a demonstration of quantum coherence (history model, Theorem 2).

If every positive-probability arbitrarily small joint position–momentum ball can be read without disturbance and its conditional law is an admissible preparation, then the covariance-action infimum is zero. Weak closure additionally admits a point state. This is an ideal classical instrument postulate, not a property proved for physical apparatus (history model, Theorem 4; see also the earlier two-pointer test).

The classical entropy-floor witness passes actual cusp transport, copy mixers and W_2 closure, but a single Gaussian position meter and a coarse outcome of probability at least $9/32$ can produce arbitrarily small posterior covariance action. The example's preparation aperture grows without bound. Gaussian inputs pass that meter test. The W_2-closed class of mixtures of admissible Gaussians repairs the meter and mixer closure, but a specified nonlinear classical canonical shear carries a member outside that class. A Wigner-positivity test also excludes the meter counterexample inputs from the quantum state space when the meter scale is identified with $\hbar/(2S_0)$. These are scoped tests, not a no-go theorem for every other classical completion (Gaussian meter, Theorems 1–3).

4. What the quantum constructions establish

For the stated smooth one-dimensional scalar potentials, a supplied $\hbar > 0$ defines a self-adjoint Hamiltonian and finite-slice products. On coherent initial packets, means on a fixed time interval have an $O(\hbar + |\pi|)$ bound and widths are $O(\sqrt{\hbar})$; any joint vanishing schedule for the parameter and mesh recovers the unique Newton solution. The $O(\hbar)$ mean rate uses bounded third derivatives. This is an observable estimate, not uniform propagator or wavefunction convergence (construction, §5; groupoid state limit).

A conservative global-Lipschitz force certificate also supports a noncommutative quantum lift with a supplied positive action parameter and an $O(\sqrt{\hbar})$ packet comparison, using the stated operator toolkit. This weaker rate is sharp in the specified $C^{1,1}$ example. The assertion that *every* classical existence proof contains a quantum construction remains a conjecture; a classical Koopman unitary group alone does not supply noncommuting observables (bounded lift, conjecture).

For the bounded cusp, two quantum Gaussian preparation families have the same limiting incoming history but different outgoing branches; the limit does not select a unique classical continuation without specifying the preparation (quantum cusp). For the specified free half-line packet, the Dirichlet boundary-force measure converges to the elastic impulse $2|p_0|\delta_{t_*}$; Neumann stress has the same limit through a different term (wall impulse). The separate relativistic extension gives a one-particle, positive-energy joint limit, not an interacting Lorentz-covariant field theory.

At fixed $\hbar$, both cell orders converge to the same specified quantum Hamiltonian. Along some joint action/mesh limits their rays can remain separated even when both means reach the same classical point; factor ordering is another, distinct ambiguity. The corresponding Lean items are specifications, not compiled quantum proofs (verification and ordering, §11). The quartic $0 + 1$ construction assembles standard Euclidean and spectral ingredients into a positive-gap model at supplied $\hbar$; it does not prove four-dimensional Yang–Mills or the necessity of that input.

The unifying open issue is therefore physical: which independently motivated preparation, continuity, record and composition laws are actually mandatory, and do they rule out the classical alternatives or specify genuinely quantum structure and a universal nonzero action? The dependency paths and obstruction map route the competing premises and their countertests without duplicating the proofs.

1. What must exist, and in what sense?

For a particle in $\mathbb{R}^d$, mass $m > 0$ and conservative potential V, Newton's initial-value problem is

$$\dot{q} = v, \qquad m\dot{v} = -\nabla V(q), \qquad (q(0), v(0)) = (q_0, v_0). \tag{1}$$

It asks for a **trajectory** from initial position and velocity. The endpoints of a path integral ask a different question: prescribe $q(0)$ and $q(T)$ and integrate over intermediate configurations. Even for the harmonic oscillator an endpoint map can become singular at a conjugate time while the initial-value trajectory remains unique.

For $V \in C^1$ with locally Lipschitz ∇V, (1) has a unique local solution, by the usual Cauchy–Lipschitz theorem. Along it,

$$E(q,v) = \tfrac{1}{2}m|v|^2 + V(q), \qquad \frac{d}{dt}E = 0. \tag{2}$$

If V is bounded below, (2) bounds $|v|$. On any finite interval $|q(t)| \le |q_0| + |t| \sup |v|$, so (q,v) remains in a compact set. The local continuation theorem then gives a unique solution for all positive and negative times. Coercivity of V is a stronger condition than is needed for this finite-time argument.

Smoothness alone does not guarantee global existence. For $m = 1$, $V(q) = -q^4/4$, $q_0 > 0$ and $v_0 = q_0^2/\sqrt{2}$, the zero-energy solution

$$q(t) = \frac{q_0}{1 - q_0 t/\sqrt{2}} \tag{3}$$

escapes to infinity at $t = \sqrt{2}/q_0$. A force singularity can also stop a classical solution: for example, a radial attractive Kepler trajectory can reach $r = 0$ in finite time. A rule for continuing through the singular point is further data.

Regularity also matters for **uniqueness**. For $m = 1$ and $V(q) = -\frac{2}{3}|q|^{3/2}$ near zero, $-V'(q) = \operatorname{sgn}(q)\sqrt{|q|}$. With $(q_0, v_0) = (0,0)$ there is the stationary solution and, for every $t_0 \ge 0$, the solution that waits at zero until t_0 and then follows $q(t) = (t - t_0)^4/144$, as long as it stays in the displayed neighbourhood. The potential can be cut off smoothly away from zero and shifted by a constant to be nonnegative and bounded. Its quantum Hamiltonian is then a bounded perturbation of the free self-adjoint Hamiltonian, while its classical initial-value problem is nonunique. Quantum existence by itself therefore does not supply a unique classical branch. The later cusp-selection calculation constructs two quantum preparations with the same limiting incoming phase-space history but different outgoing limits.

2. A velocity-dependent force: magnetism

In three dimensions the Lorentz force, with a static scalar potential and a static magnetic field, gives

$$\dot{q} = v, \qquad m\dot{v} = -\nabla V(q) + e\, v \times B(q). \tag{4}$$

It is not a force depending on position alone. If ∇V and B are defined and locally Lipschitz on all of $\mathbb{R}^3$, the full phase-space vector field is locally Lipschitz, so local existence and uniqueness still hold. Moreover $v \cdot (v \times B) = 0$, hence the energy (2) is conserved. If V is bounded below, the same speed and position bounds as above prove global existence and uniqueness, even if B is not globally bounded.

For instance, if V is constant and $B = B_0\hat{z}$, the field rotates the transverse velocity with cyclotron frequency eB_0/m and leaves the longitudinal velocity constant. A velocity-dependent force is thus not an exception to the ODE theory; its dependence must be included in the phase-space vector field being estimated.

If $B = \nabla \times A$ on $\mathbb{R}^3$, the quantum Hamiltonian is

$$H_{\hbar,A} = \frac{1}{2m}(-i\hbar\nabla - eA(q))^2 + V(q). \tag{5}$$

The kinetic momenta generally do not commute: $[\pi_i, \pi_j] = i\hbar e(\partial_i A_j - \partial_j A_i)$. The positive Brownian-bridge density below does not directly cover this magnetic problem: a magnetic Euclidean representation contains a phase. The present coherent-packet theorem is proved for the scalar Hamiltonian in (11), not for (5).

3. A wall is an impact rule, not a smooth force

Take the half-line $q \geq 0$ and a free particle starting at $q_0 > 0$ with $p_0 < 0$. For an ideal **elastic** wall, the prescribed impact law is $p(t_*+) = -p(t_*-)$ at $t_* = -mq_0/p_0$. The motion is

$$q(t) = |q_0 + p_0 t/m|, \qquad p(t) = p_0 \operatorname{sgn}(q_0 + p_0 t/m) \quad (t \neq t_*). \tag{6}$$

Position is continuous; momentum has a jump. Equation (1) with a finite smooth force does not describe that jump. Nonpenetration alone does not select the elastic law: a restitution coefficient or another contact rule would be additional data.

For a smooth wall in several dimensions, the elastic specular rule reverses the normal component and retains the tangential component: $v_+ = v_- - 2(v_- \cdot n)n$. This requires a specified unit normal and a well-defined collision. Corners, grazing contacts and an accumulation of impacts require separate analysis; one should not infer their uniqueness merely from the local smooth-force theorem.

A mathematically defined quantum impenetrable wall can be represented by the Dirichlet Laplacian

$$H_{\hbar,D} = -\frac{\hbar^2}{2m}\frac{d^2}{dq^2}, \qquad D(H_{\hbar,D}) = H^2(0,\infty) \cap H_0^1(0,\infty). \tag{7}$$

It is self-adjoint and gives a unitary group. Its Euclidean kernel is

$$K_D(T;x,y) = k_{\hbar,T}(x-y) - k_{\hbar,T}(x+y), \quad k_{\hbar,T}(r) = \sqrt{\frac{m}{2\pi\hbar T}}e^{-mr^2/(2\hbar T)}, \quad x,y>0. \tag{8}$$

This is the method of images: the Euclidean reference process is Brownian motion killed at the boundary. The quantum real-time wall reflects, but killed Euclidean paths should not themselves be read as real-time bouncing trajectories. Section 8 proves a quantitative coherent-packet position limit through the bounce.

A subtle domain issue is important: the symmetric differential operator $-i\hbar\, d/dq$ on the half-line with Dirichlet boundary has unequal deficiency indices and no self-adjoint extension on $L^2(0,\infty)$. The Hamiltonian (7) is nevertheless self-adjoint. A current or a quadratic-form expectation involving $-i\hbar d/dq$ can be studied, but it should not be presented as the expectation of a self-adjoint half-line momentum operator. See Al-Hashimi and Wiese, arXiv:2103.01715 (abstract) for the boundary momentum problem.

4. A finite Euclidean lattice and one exact block

Return to the whole line and choose endpoints x, y, duration $T = u + v$, and action unit $\hbar > 0$. The free Euclidean kernel is $k_{\hbar,T}$ from (8). Its semigroup identity, refined at time u, is

$$k_{\hbar,T}(y-x) = \int_{\mathbb{R}} k_{\hbar,u}(z-x)k_{\hbar,v}(y-z)\,dz. \tag{9}$$

Completing the square gives the **conditional** law of the intermediate coordinate:

$$k_{\hbar,u}(z-x)k_{\hbar,v}(y-z) = k_{\hbar,T}(y-x)\sqrt{\frac{a}{2\pi\hbar}}\,e^{-a(z-\mu)^2/(2\hbar)}, \quad a = \frac{mT}{uv}, \quad \mu = \frac{vx+uy}{T}. \tag{10}$$

The normalization in (10) matters: an action penalty is not yet a probability bound until the full conditional integral is divided out. The inserted point has mean μ and variance $\hbar/a = \hbar uv/(mT)$.

For a nonnegative potential, the finite-time weight adds $\exp[-\sum_j \delta_j(V(q_j)+V(q_{j+1}))/(2\hbar)]$. As the mesh tends to zero, the Brownian-bridge expectation converges by dominated convergence. With $Z_\hbar(x,y)$ denoting that expectation, the kernel of the Euclidean semigroup is

$$K_\hbar(T;x,y) = k_{\hbar,T}(y-x)Z_\hbar(x,y), \qquad H_\hbar = P^2/(2m) + V(Q). \tag{11}$$

Feynman–Kac and the spectral theorem then provide the real-time $U_\hbar(t) = e^{-itH_\hbar/\hbar}$. The Brownian measure and these standard operator theorems are inputs; the construction is not a lattice-only derivation of Wiener measure.

There is also a sign distinction. The Euclidean stationary equation for $\int(\frac{1}{2}m\dot{q}^2 + V)\,dt$ is $m\ddot{q} = +V'(q)$. The real-time equation for $\int(\frac{1}{2}m\dot{q}^2 - V)\,dt$ is $m\ddot{q} = -V'(q)$. The positive-weight minimizer cannot be relabelled as a Newton trajectory without another argument.

5. Reading Dimock and Bałaban from this example

Dimock's two papers develop Bałaban's method for a **three-dimensional lattice scalar ϕ^4 theory**, with ultraviolet stability of its partition function as the illustrative problem. Their scalar results do not by themselves establish the Yang–Mills mass gap. The following references are to specific statements and passages, not a claim that their full proofs have been audited.

1. **Keep the complete density.** In Dimock I, §2.1, equations (15), (26)–(28), Q averages a fine field over a block; the next density is obtained by integrating the previous density against a normalized Gaussian kernel, then rescaling. Equation (28) preserves the total integral. The $0+1$ operation (10) instead eliminates a time coordinate conditional on retained coordinates. Our density $\rho_n = \mathbb{E}_0[\rho \mid \mathcal{F}_n]$ is relative to a Brownian-bridge reference and satisfies $\rho_n = \mathbb{E}_0[\rho_{n+1} \mid \mathcal{F}_n]$. This is an exact tower, but it is not literally Dimock's noisy block-average map, nor does it contain his rescaling.

2. **A minimizer is only the center of an integral.** Completing the free square in (10) produces a background point μ *and* its covariance $\hbar/a$ and normalization. Dimock I, §§2.2–2.3, likewise separates a background and Gaussian fluctuation in its free flow. With interactions, the analogue is a full conditional law around a generally nonlinear minimizer, not merely the minimizer itself.

3. **Generated interactions must travel to the next scale.** For one inserted vertex, with $w = T/2$, define

$$A_\hbar(\mu) = -\hbar \log\left[\sqrt{\frac{a}{2\pi\hbar}} \int e^{-[a(z-\mu)^2/2 + wV(z)]/\hbar} dz\right]. \tag{12}$$

   This entire function enters the retained kernel. If $V(z) = \lambda z^4/4$, differentiating at $\lambda = 0+$ gives

$$\partial_\lambda A_\hbar(\mu)|_{0+} = \frac{w}{4}\left[\mu^4 + 6\mu^2\frac{\hbar}{a} + 3\left(\frac{\hbar}{a}\right)^2\right]. \tag{13}$$

   Even one integration generates a quadratic term and a constant in addition to the quartic term. The derivative is not a claim of convergence of the whole perturbation series. Dimock I, §4.1, equations (176)–(180), and Theorem 14 propagate a much richer representation: a normalization, mass and energy terms, a quartic term and a controlled polymer remainder, under the theorem's small-coupling and sufficiently-large-block hypotheses. The nonlinear-cell calculation gives nonperturbative bounds for (12).

4. **Small fields are not the whole integral.** Dimock I, equations (172)–(175), introduces characteristic functions for the small-field analysis. Dimock II, introduction and §3.3, restores the full expansion by summing over histories of regions; its Theorem 3.1 includes large-field and boundary terms and normalization, under its stated block-size, mass and running-coupling conditions and with the energy and mass counterterms selected in Part I. The elementary identity $\prod_b[\chi_b + (1-\chi_b)] = 1$ illustrates why every pattern of good and bad blocks is retained. Dimock's good-region tests involve fields, derivatives and nested

regions; they are not identical to a bound on one midpoint fluctuation. In the one-cell law, strong convexity yields the normalized bound

$$\mathbb{P}(|Z - z_*| \geq r \mid \text{retained endpoints}) \leq 2e^{-\kappa r^2/(2\hbar)}, \tag{14}$$

provided the conditional potential has second derivative at least $\kappa > 0$. The integration over its complement is controlled, not deleted.

5. **Locality of correlated fluctuations must be earned.** Dimock I, §2.4 and Lemma 6, constructs a Green function using local inverses and a convergent random-walk expansion; §4.5 introduces weakening parameters before the cluster expansion. Dimock II, §3.13 and Lemmas 3.15–3.17, localizes the interaction, remainder and existing boundary terms; §3.14 and Lemma 3.18 then perform a cluster expansion with holes. Locality in a correlated fluctuation field is not automatically locality in independent Gaussian variables. In the nearest-neighbour time chain, nonadjacent eliminated vertices are conditionally independent **for one halving**. That elementary fact does not prove a multi-scale polymer estimate in three or four dimensions.

The continuity of the normalized 0+1 density tower follows here from conditional expectation and an already constructed Brownian bridge. The difficult field-theory task is to retain useful **uniform, local bounds** on effective interactions, large-field regions and boundaries as the cutoff is removed. An exact composition identity alone is not that estimate.

6. From the quantum blocks to a Newton path

Fix a Gaussian coherent packet centered at (q_0, p_0) with position and momentum standard deviations of order $\sqrt{\hbar}$. At each time slice first kick by the potential and then drift freely:

$$K_{\hbar,s} = e^{-isV(Q)/\hbar}, \qquad D_{\hbar,s} = e^{-isP^2/(2m\hbar)}, \qquad T_{\hbar,s} = D_{\hbar,s}K_{\hbar,s}. \tag{15}$$

For finite products the exact conjugations are

$$K^*_{\hbar,s} P K_{\hbar,s} = P - sV'(Q), \qquad D^*_{\hbar,s} Q D_{\hbar,s} = Q + sP/m. \tag{16}$$

If $|V''| \leq L$, the standard deviation of $V'(Q)$ is at most L sd(Q). A discrete Gronwall estimate consequently bounds both position and momentum spreads by $C_T\sqrt{\hbar}$, uniformly over partitions of $[0, T]$.

Let $z_{\hbar,\pi}(t)$ interpolate the means after the finite cells. Under the detailed theorem's bounded V''' assumption, expand V' about the packet's mean. The linear term averages to zero, and the remaining force bias is bounded by $\frac{1}{2}\|V'''\|_\infty \operatorname{Var}(Q) = O(\hbar)$. Together with the spread bound and (16), this gives the force-balance residual

$$\|\dot{z}_{\hbar,\pi} - \mathcal{F}(z_{\hbar,\pi})\| \leq C_T(\hbar + |\pi|), \qquad \mathcal{F}(q,p) = (p/m, -V'(q)). \tag{17}$$

Comparing two independently chosen $\hbar$'s and meshes, and applying Gronwall, shows that these interpolants form a Cauchy family. Its limit is the unique solution of (1), with

$$\sup_{t \leq T} \|z_{\hbar,\pi}(t) - z(t)\| \leq C_T(\hbar + |\pi|). \tag{18}$$

The detailed proof gives the weighted phase-space norm, constants, arbitrary-partition product limit and domains. The packet's width still scales as $O(\sqrt{\hbar})$; the sharper estimate concerns its means. It does not assert that finite-product propagators converge uniformly in operator norm along arbitrary joint schedules.

Cauchy–Lipschitz constructs the integral equation directly, usually by a contraction on a short interval. The route above constructs quantum operators first and their classical limit

afterward. Both use the same Lipschitz stability of the force and Gronwall uniqueness. The extra quantum structure relates the solution to a family of states; it does not weaken the regularity needed here or derive $\hbar > 0$ from Newton's law. Hepp's 1974 coherent-state theorem is established prior work on quantum correlation functions and classical orbits (abstract and introduction read).

7. The tangent groupoid and the separate scales

For $M = \mathbb{R}$, the tangent groupoid has positive fibers $M \times M$ and a zero fiber TM. Its observable C^*-algebra is a continuous field, with

$$\mathcal{A}_{\epsilon>0} \simeq \mathcal{K}(L^2(\mathbb{R})), \qquad \mathcal{A}_0 \simeq C_0(T^*\mathbb{R}). \tag{19}$$

After choosing an action unit S_* and identifying $\hbar = \epsilon S_*$, the state estimate in the groupoid companion connects the coherent quantum states to evaluation on the constructed classical trajectory. This uses the established continuous-field and strict quantization framework of Cariñena et al. and Landsman–Ramazan (abstracts read).

Three operations remain distinguishable: ϵ labels a geometric deformation; the time mesh controls a finite product; and blocking integrates over intermediate variables. Choosing a joint schedule ties their parameters, but does not make geometric zoom identical to Wilsonian integration or prove that the physical action unit runs.

The subsequent dispersion construction extends the packet argument to the positive-energy relativistic Hamiltonian and controls its nonrelativistic limit. Its free spectral Dirichlet wall is a separate nonlocal boundary model.

8. A quantum packet through the elastic bounce

The wall example can be related to quantum theory without applying the smooth-force theorem at $q = 0$. Let $q_0 > 0$, $p_0 < 0$, $\omega > 0$, and let $g_\hbar$ be the normalized full-line Gaussian of position mean q_0, momentum mean p_0, and variances $\hbar/(2m\omega)$ and $\hbar m\omega/2$. If R is reflection, $(Rg)(x) = g(-x)$, define the normalized odd state

$$\phi_\hbar = \frac{g_\hbar - Rg_\hbar}{\sqrt{2(1-\gamma_\hbar)}}, \qquad \gamma_\hbar = \langle g_\hbar, Rg_\hbar\rangle = \exp\left[-\frac{m\omega q_0^2 + p_0^2/(m\omega)}{\hbar}\right]. \tag{20}$$

The corresponding Dirichlet half-line state is $\psi_\hbar(q) = \sqrt{2}\,\phi_\hbar(q)$ for $q > 0$. Odd extension intertwines its evolution with free evolution on the full line.

Write $a(t) = q_0 + p_0 t/m$, and let $g_{\hbar,t}$ be the freely evolved Gaussian. Its position variance is

$$\sigma_q(t)^2 = \frac{\hbar}{2m\omega}(1 + \omega^2 t^2). \tag{21}$$

Because $||x| - |a|| \le |x - a|$ and reflection commutes with the free evolution,

$$\left|\langle Q\rangle_{\psi_\hbar(t)} - |a(t)|\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\,\sigma_q(t). \tag{22}$$

Indeed the half-line position expectation equals the full-line expectation of $|X|$ in the odd state; apply the triangle inequality to the two Gaussian terms. Thus position converges uniformly on every bounded time interval, **including the impact time**, to the elastic classical trajectory (6).

Away from the impact, the corresponding probability-current momentum also converges to the incoming or reflected value. More precisely, if $|a(t)| \ge \eta > 0$, with $\sigma_p = \sqrt{\hbar m\omega/2}$,

$$\left|\int_0^\infty \overline{\psi_\hbar}(-i\hbar\partial_q)\psi_\hbar\,dq - p_0 \operatorname{sgn} a(t)\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\left(\sigma_p + \frac{2|p_0|\sigma_q(t)}{\eta}\right). \tag{23}$$

For the proof, the half-line integral is the full-line expectation of $\operatorname{sgn}(X)P$ in the odd state. On the Gaussian, $\|(\operatorname{sgn} X\, P - \operatorname{sgn} a\, p_0)g_t\|$ is at most $\sigma_p + 2|p_0|\mathbb{P}(aX < 0)^{1/2}$; Chebyshev bounds the last probability by $\sigma_q(t)^2/a(t)^2$. The reflected Gaussian obeys the same estimate. For each fixed $T, \eta > 0$ independent of $\hbar$, (23) proves uniform convergence over $|t| \le T$ with $|a(t)| \ge \eta$. It gives no uniform bound in an $\hbar$-dependent impact layer $|a(t)| \sim \sqrt{\hbar}$ or momentum value at t_*. The expression in (23) is a real current form on this domain, not an expectation of a self-adjoint half-line momentum.

The bound deliberately excludes t_*. A family of continuous quantum current expectations cannot converge uniformly across t_* to the discontinuous classical momentum in (6). Position convergence and momentum convergence therefore have different valid topologies. The separate wall-impulse calculation resolves this impact layer: the Dirichlet boundary-force measures converge to $2|p_0|\delta_{t_*}$, while the full Neumann boundary stress has the same limit even though its boundary derivative vanishes.

9. Consequence for STATE

The recent synthesis at the start consolidates the October 27–28 proof records and their classical countertests into the live Newton entry point. It keeps the physical premises, the comparison with quantum state positivity, and the universal nonzero-action question separate.

The requested didactic bridge now includes explicit source locations, the normalized one-cell operation, generated terms and the limitation of the analogy. It distinguishes classical smooth-force existence, continuation, nonuniqueness, magnetic forces and an imposed hard-wall impact. The wall packet calculation gives a separate rigorous example of a classical limit through an impact. The supplied action parameter, the independent action-necessity problem and the higher-dimensional field-theory estimates remain open in their respective scopes. The later canonical central-limit calculation identifies a Gaussian preparation sector under additional copying and cusp-compatibility assumptions; its floor is not identified with the quantum phase constant.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). The impact packet estimates and their convergence domains, the magnetic continuation argument, and the cited Dimock source locations were checked against the written calculation and selected source passages. This is an internal model check, not independent peer review of the imported ODE, operator, probability or field-theory theorems. The linked later cusp-selection and wall-impulse notes are separate written derivations and were not part of that check.